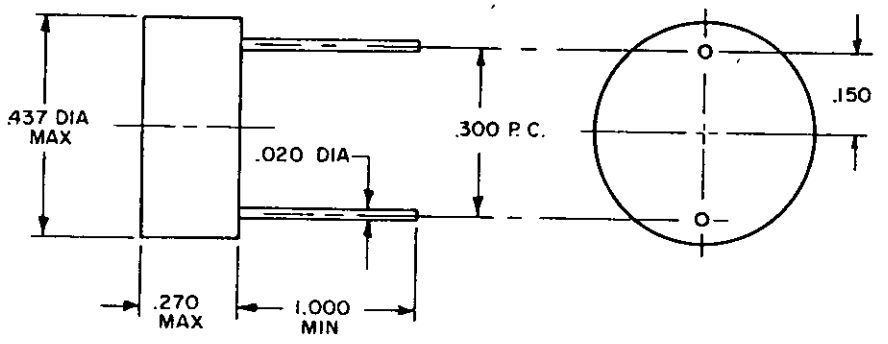
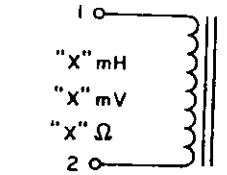
MILITARY SPECIFICATION SHEET

TRANSFORMERS AND INDUCTORS (AUDIO, POWER, AND HIGH-POWER PULSE) INDUCTORS, AUDIO FREQUENCY, TF5S20ZZ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the inductors described herein shall consist of this document and the latest issue of Specification MIL-T-27.





WV-250 V
ALTITUDE - 70,000 FT MAX
(SEE TABLE I FOR VALUE OF "X")

CIRCUIT DIAGRAM AND MARKING

INCHES	MM
.010	.25
.020	.51
.150	3.81
.270	6.86
.300	7.62
.437	11.10
1.000	25.40

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Tolerance on terminal position dimensions is ±.010.
4. Marking shall be on the side of the case opposite terminals.
5. Electrical values shall be marked as specified in table I, as applicable.

FIGURE 1. Dimensions and configuration.

FSC 5950

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REQUIREMENTS: (When numbers in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings: See table I.
Working voltage: 250 volts.

Design and construction:
Dimensions and configuration: See figure 1.
Duty cycle: Continuous.
Case: Encapsulated.
Material: Epoxy.
Altitude: 70,000 feet, maximum.
Terminals: Tinned copper, 24 AWG.
Terminal length: 1 inch minimum.
Weight: 2 grams, maximum.
Operating temperature range: -55° to +130°C.

Terminal strength: MIL-STD-202, method 211, test condition A, 2 pounds.

Dielectric withstanding voltage:
At sea level: 1,000 Vrms.
At reduced barometric pressure: 300 Vrms.

Electrical characteristics (see table I): Power level 0.5 watt maximum.

Temperature rise: 40°C at an ambient temperature of 90°C.

Vibration (high frequency): MIL-STD-202, method 204, test condition D.

Marking location: See figure 1.

Part number: M27/214-(dash number from table I).

Supersession data:
MIL-T-27/214 supersedes MIL-T-27/94(USAF) which is inactive for design after the date of this specification sheet.

Superseding part numbers:
M27/214-08, M27/214-11 and M27/214-14 supersedes M27/94-01, M27/94-02 and M27/94-03, respectively.

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TABLE I. Electrical ratings. 1/

Dash No.	Inductance (mH) (1-2)	Voltage (mV) (1-2)	Self-resonant frequency (MHz) (min)	DC resistance (ohms) (1-2) (max)	Temperature coefficient (-55° to +130°C)	Quality factor (min)	Distrib. capacitance (pF) (max)
01	0.05±2%	11.3	9.000	1.89	±50 PPM	37 at 300 kHz	6
02	0.10±2%	16.0	6.000	2.64	±50 PPM	45 at 250 kHz	7
03	0.33±2%	29.2	2.800	7.20	±50 PPM	45 at 250 kHz	10
04	0.47±2%	34.9	2.300	10.00	±50 PPM	45 at 250 kHz	10
05	0.47±2%	13.8	2.200	2.30	±1%	20 at 20 kHz	11
06	1.00±2%	50.6	1.300	24.50	±50 PPM	23 at 400 kHz	15
07	1.00±2%	20.1	1.100	3.94	±1%	30 at 20 kHz	21
08	1.00±2%	13.7	1.800	2.30	±1%	40 at 100 kHz	6
09	3.30±1%	92.2	0.630	76.00	±50 PPM	23 at 100 kHz	19
10	3.30±1%	36.7	0.720	9.00	±1%	30 at 20 kHz	15
11	3.30±1%	25.0	0.860	5.50	±1%	40 at 100 kHz	9
12	10.00±1%	160.0	0.310	239.00	±50 PPM	23 at 100 kHz	26
13	10.00±1%	43.8	0.390	17.10	±1%	27 at 8 kHz	17
14	10.00±1%	65.0	0.475	26.00	±0.25%	60 at 100 kHz	15
15	33.00±1%	116.0	0.160	114.00	±1%	28 at 20 kHz	30
16	33.00±1%	74.9	0.190	50.20	±1%	14 at 5 kHz	21
17	100.00±1%	201.2	0.060	396.00	±1%	14 at 10 kHz	70
18	100.00±1%	138.6	0.090	207.00	±1%	14 at 5 kHz	31
19	250.00±1%	219.0	0.049	435.00	±1%	14 at 5 kHz	42

1/ Complete qualification testing and approval to M27/214-19 shall be sufficient to grant qualification approval to M27/214-01 through M27/214-18.

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Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - SH, OS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AR, WC
Navy - MC, AS
Air Force - 19

Preparing activity:
Army - EL

Agent:
DLA - ES

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